Class 10 Task Program:- ADC PWM Modular Programming

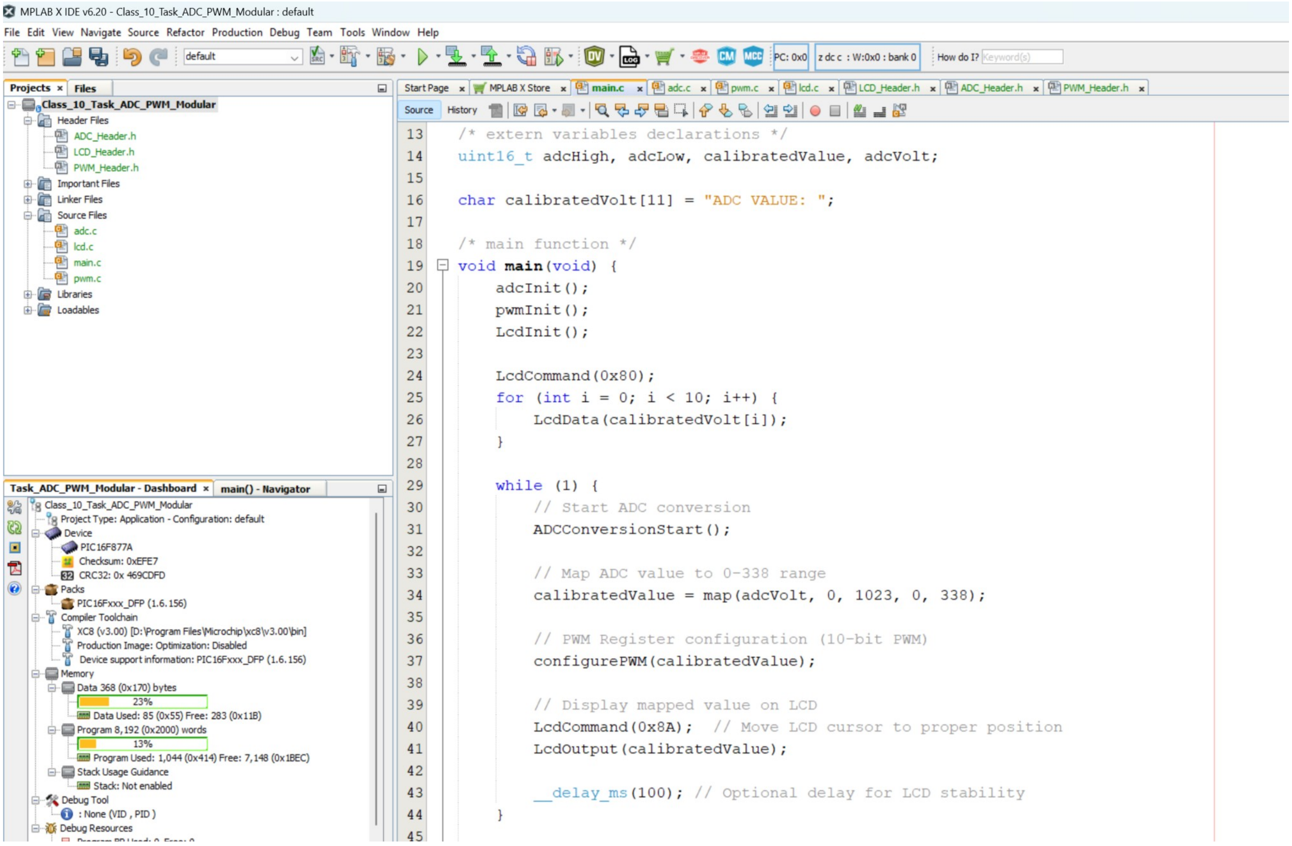


Figure 1: modular Programming Setup

```c
/*
 * File:   main.c
 * Author: sagar
 *
 * Created on 2 June, 2025, 10:37 PM
 */
#include <xc.h>
#include "ADC_Header.h"
#include "LCD_Header.h"
#include "PWM_Header.h"
#define  _XTAL_FREQ 6000000

/* extern variables declarations */
uint16_t adcHigh, adcLow, calibratedValue, adcVolt;

char calibratedVolt[11] = "ADC VALUE: ";

/* main function */
void main(void) {
    adcInit();
    pwmInit();
    LcdInit();

    LcdCommand(0x80);
    for (int i = 0; i < 10; i++) {
        LcdData(calibratedVolt[i]);
    }

    while (1) {
        // Start ADC conversion
        ADCConversionStart();

        // Map ADC value to 0-338 range
        calibratedValue = map(adcVolt, 0, 1023, 0, 338);

        // PWM Register configuration (10-bit PWM)
        configurePWM(calibratedValue);

        // Display mapped value on LCD
        LcdCommand(0x8A);  // Move LCD cursor to proper position
        LcdOutput(calibratedValue);

        __delay_ms(100); // Optional delay for LCD stability
    }

    return;
}
```

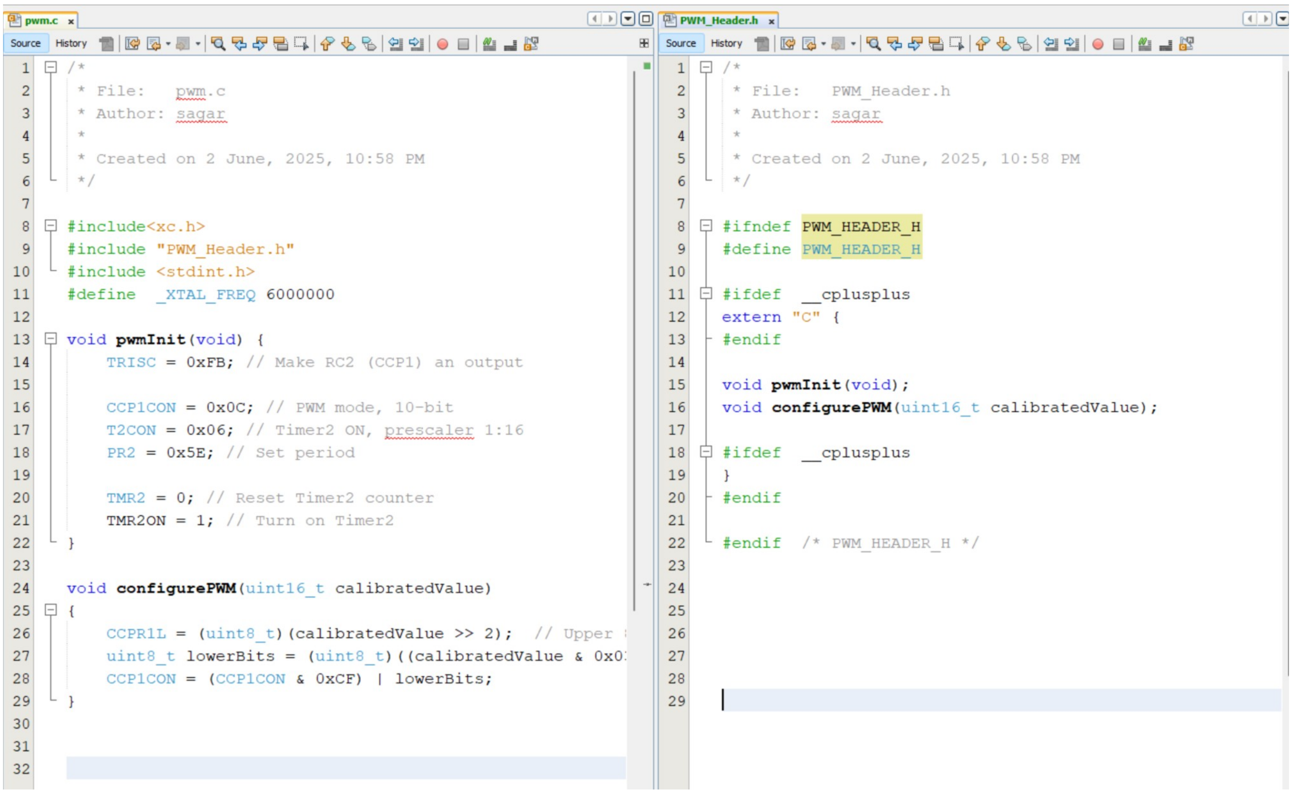


Figure 2: pwm;- .c .h

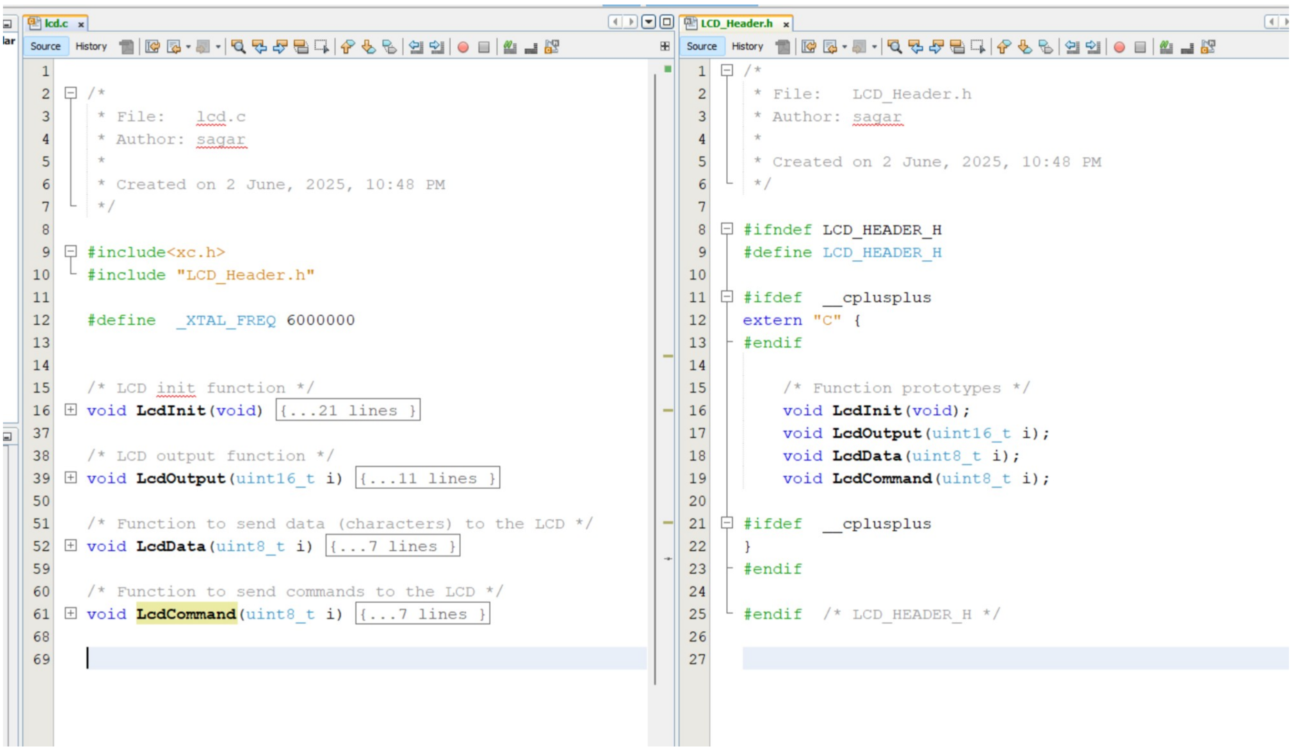
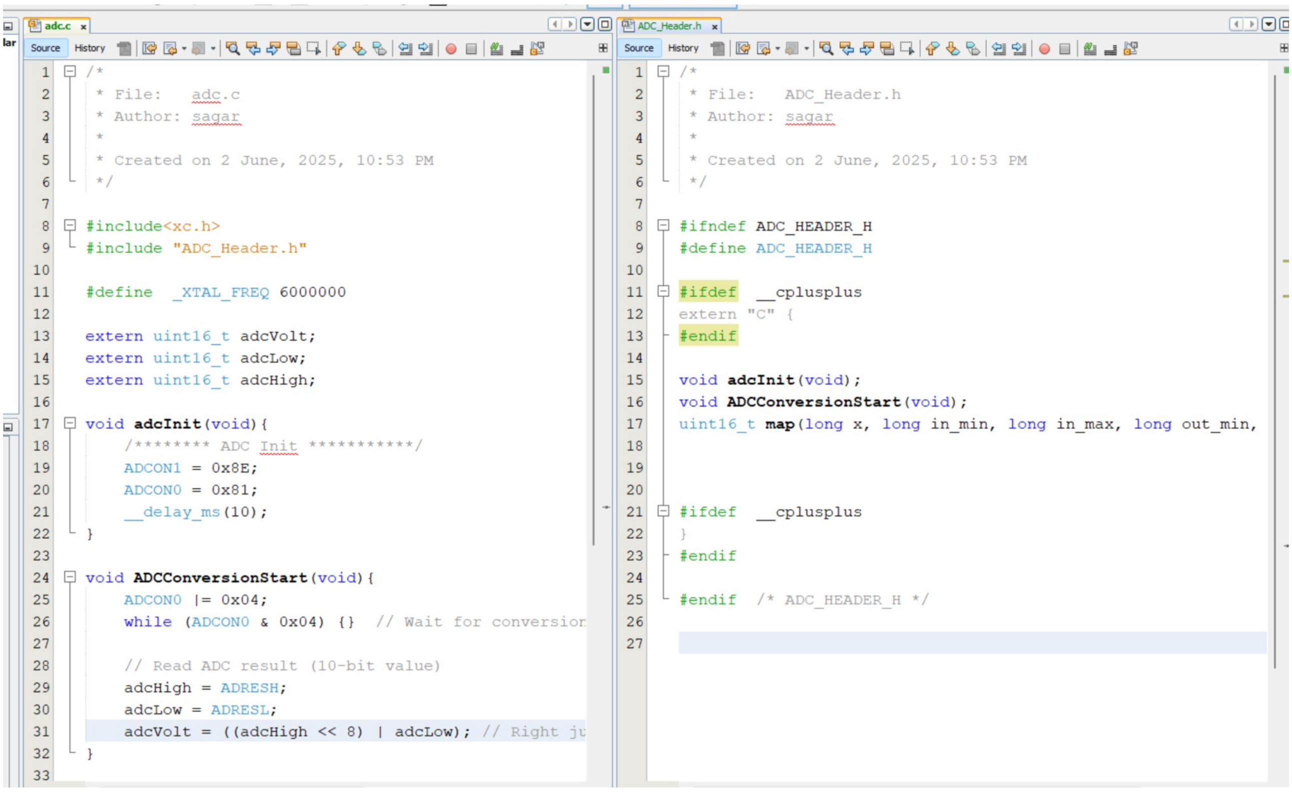


Figure 3: LCD:- .c .h



```c
/*
 * File:   adc.c
 * Author: sagar
 *
 * Created on 2 June, 2025, 10:53 PM
 */

#include<xc.h>
#include "ADC_Header.h"

#define  _XTAL_FREQ 6000000

extern uint16_t adcVolt;
extern uint16_t adcLow;
extern uint16_t adcHigh;

void adcInit(void){
    /******** ADC Init ***********/
    ADCON1 = 0x8E;
    ADCON0 = 0x81;
    __delay_ms(10);
}

void ADCConversionStart(void){
    ADCON0 |= 0x04;
    while (ADCON0 & 0x04) {}  // Wait for conversion

    // Read ADC result (10-bit value)
    adcHigh = ADRESH;
    adcLow = ADRESL;
    adcVolt = ((adcHigh << 8) | adcLow); // Right ju
}
```

```c
/*
 * File:   ADC_Header.h
 * Author: sagar
 *
 * Created on 2 June, 2025, 10:53 PM
 */

#ifndef ADC_HEADER_H
#define ADC_HEADER_H

#ifdef  __cplusplus
extern "C" {
#endif

void adcInit(void);
void ADCConversionStart(void);
uint16_t map(long x, long in_min, long in_max, long out_min,



#ifdef  __cplusplus
}
#endif

#endif  /* ADC_HEADER_H */
```

Figure 4: ADC:- .c .h